Thermal Control & Heat Rejection: System Design Description

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Revision	B
Status	APPROVED
Classification	INTERNAL
Author	Soren Kessel
Approved by	Aldric Voss
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1. System overview

Thermal control carries surplus reactor and habitat heat to the radiator fin arrays on the station surface through heat exchanger stacks, pumped coolant loops and passive heat pipes.

The system is led by Reactor & Power Systems. It is delivered through 6 work packages with a combined budget of 74,409.65 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
TX-HXS	Heat exchanger stack	SHV-TX-SPEC-0101	SHV-TX-DWG-0201	SHV-TX-BOM-0301
TX-RFA	Radiator fin array	SHV-TX-SPEC-0007	SHV-TX-DWG-0202	SHV-TX-BOM-0302
TX-CPS	Coolant pump station	SHV-TX-SPEC-0103	SHV-TX-DWG-0203	SHV-TX-BOM-0303
TX-HPL	Heat pipe loop	SHV-TX-SPEC-0104	SHV-TX-DWG-0204	SHV-TX-BOM-0304

3. Architecture

3.1 Heat exchanger stack

The heat exchanger stack is specified in SHV-TX-SPEC-0101. Its governing parameter is the rejection capacity, set at 0.66 PW. The responsible engineer is Soren Kessel.

3.2 Radiator fin array

The radiator fin array is specified in SHV-TX-SPEC-0007. Its governing parameter is the fin area, set at 6.3 km². The responsible engineer is Dalric Torvald.

3.3 Coolant pump station

The coolant pump station is specified in SHV-TX-SPEC-0103. Its governing parameter is the pump flow, set at 104 m³/s. The responsible engineer is Rhett Starke.

3.4 Heat pipe loop

The heat pipe loop is specified in SHV-TX-SPEC-0104. Its governing parameter is the loop length, set at 5.78 km. The responsible engineer is Fallon Darrow.

4. Interfaces

Document	Title
SHV-ICD-HR-TX	Interface Control Document: Hypermatter Reactor / Thermal Control & Heat Rejection

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-TX-01	Thermal Control & Heat Rejection: Design & analysis	Emrys Bresk	COMPLETE	100.0%
WP-SHV-TX-02	Thermal Control & Heat Rejection: Fabricate & install heat exchanger stack	Emrys Bresk	IN_PROGRESS	86.8%
WP-SHV-TX-03	Thermal Control & Heat Rejection: Fabricate & install radiator fin array	Jessa Pell	IN_PROGRESS	95.4%
WP-SHV-TX-04	Thermal Control & Heat Rejection: Fabricate & install coolant pump station	Tamsin Quarrel	IN_PROGRESS	83.0%
WP-SHV-TX-05	Thermal Control & Heat Rejection: Fabricate & install heat pipe loop	Emrys Bresk	COMPLETE	100.0%
WP-SHV-TX-06	Thermal Control & Heat Rejection: Integration & test	Alira Aster	IN_PROGRESS	89.0%

6. Risks

- Supply of heat exchanger stack assemblies depends on a single supplier.
- Supply of radiator fin array assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.